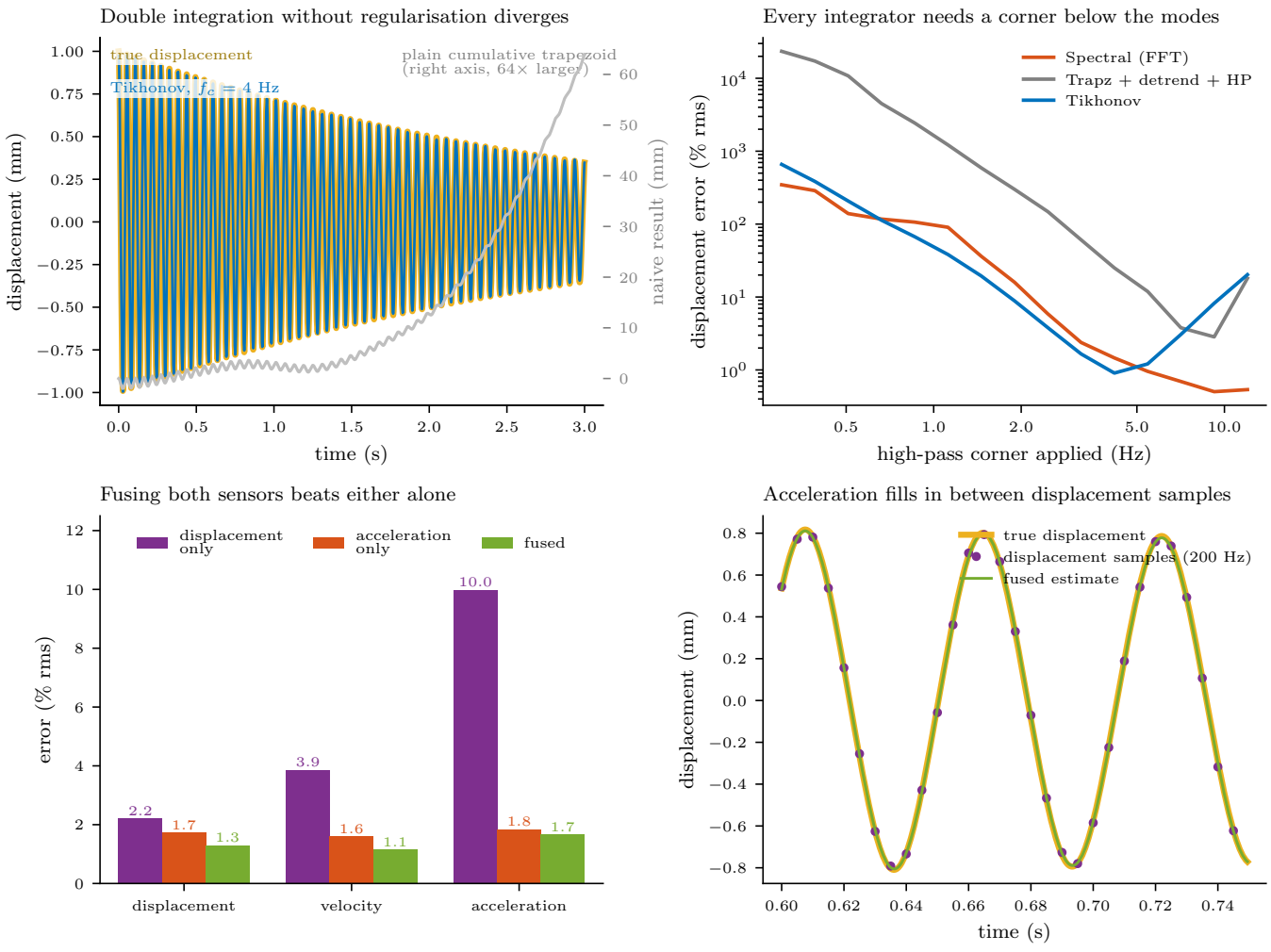

Double integration without regularisation diverges
true displacement
Tikhonov, $f_c = 4$ Hz
plain cumulative trapezoid
(right axis, 64× larger)
displacement (mm)
naive result (mm)
time (s)
Every integrator needs a corner below the modes
Spectral (FFT)
Trapz + detrend + HP
Tikhonov
displacement error (% rms)
high-pass corner applied (Hz)
Fusing both sensors beats either alone
displacement only
acceleration only
fused
error (% rms)
2.2
1.7
1.3
3.9
1.6
1.1
10.0
1.8
1.7
displacement
velocity
acceleration
Acceleration fills in between displacement samples
true displacement
displacement samples (200 Hz)
fused estimate
displacement (mm)
time (s)